Biostatistics-Summay

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1 Summary

The high-dimensional two-sample distribution test, which aims to determine whether two sets of high-dimensional samples are drawn from the same distribution, is a fundamental problem in nonparametric statistics and machine learning.

Early methods primarily include maximum mean discrepancy (MMD)[5] based on kernel embeddings, energy distance (ED)[13] based on pairwise sample distances[12], as well as graph-based[3] and nearest-neighbor-based test approaches.

These methods exhibit favorable applicability in general distribution difference detection, but are prone to power loss as dimensionality increases, particularly under weak signals or sparse alternatives where differences are concentrated in only a handful of coordinates.[10][4][17][11]

In recent years, Wasserstein distance (WD)[14][9] has garnered significant attention for its ability to capture the geometric differences between distributions, yet the naive empirical Wasserstein distance suffers from a severe curse of dimensionality in high-dimensional settings.[7]

To this end, researchers have proposed a series of variants including sliced Wasserstein distance (SWD)[1], smoothed Wasserstein distance[8], and projected Wasserstein distance (PWD)[15],kernel projected Wasserstein distance (KRWD)[16] , which improve both statistical convergence properties and computational feasibility via smoothing or low-dimensional projection.

The max-sliced Wasserstein distance (MSWD)[6][2] research line pursued in this paper further emphasizes selecting from all possible projection directions the one that maximally amplifies the discrepancy between the two distributions, and constructs a high-dimensional two-sample test by integrating sparsity constraints with bootstrap approximation (BA).

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